Computer Science Internal Assessment

IA Project: Student Management System

Criteria E-Evaluation and Feedback

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Evaluation of Product

1. The Student Management System supports role-based authentication, which ensures individual login credentials for Admin, Teacher and Student. **Achieved**
2. A role-based access control system controls individual menu choices for the various users. Admins get full access, while teachers and students have limited access. **Achieved**
3. Unauthorised system access is protected through an authentication and authorisation process. **Achieved**
4. Teachers and administrators generate student progress reports. **Achieved**
5. Admins generate fee summary reports (including paid, outstanding, and overdue fees). **Achieved**
6. Teacher and Student records are organised using unique identifiers. **Achieved**
7. The system supports CRUD operations for each of the modules, like teachers and students. **Achieved**
8. The system supports offline functionality and is scalable to accommodate future growth in student numbers. **Achieved**
9. Individual teacher and student can only see their own information. **Achieved**
10. Valuable data (password) would be secured in the database using encryption (hash salt). **Achieved**
11. This system will be flexible to include the addition of existing subjects into the system to enable easy grading and mark management. **Partially Achieved**
12. The system will be able to assist in printing and the generation of student grade reports with the inbuilt print facility that will have detailed data on marks. **Not Achieved**

FEEDBACK FROM CLIENT

The system is efficient and effective in management and security of data because it offers a secure method of managing student records, academic data and fees and it minimizes manual work. It has a user-friendly interface that is not difficult to use, but better spacing and short onscreen instructions will make it easier. Role-based access is quite effective, but a view-only teacher role would be applicable. The student records have unique identifiers, but a more organised system would make it easier to enter student information manually. The basic CRUD operations are stable though a soft-delete option may help in eliminating unintentional loss of data. Search and validation are effective, but may have multiple criteria search and more explicit errors. Offline capability is positive, but background-synchronisation and indexing of database would enhance scaling.

RECOMMENDATION FROM CLIENT

A way of enhancing the application, additional spacing and simplified instructions can be added as part of user interface to make it easier to use. A teacher role should be added in which they will only see reports but not to make any changes. Unique identifiers to be more organised to make referencing easier. Soft-delete must be introduced so that important records are not lost accidentally and can be recovered in case they are required. Multi-criteria filtering and better validation feedback to the user could be added to search functionality. Background synchronization aids in updates being made when offline functionality is being utilized. Lastly, indexing to database queries may be applied to ensure that query rate does not slug as data stored in databases grows.

The extensibility features

The system's ability to print documents is part of the extensibility features, along with the soft delete feature to avoid unintentional data loss. These feature are to be added.

Word Count Criteria E: 286 words